HDOC Day 40

Question: Write a menu-driven C program to read a student's first, middle and last name, using separate functions and pointers where outputs are modified.

Operations: 1. Combine names 2. Find full-name length 3. Copy full name 4. Compare first and last names 5. Reverse name order 6. Count vowels, consonants, digits and spaces 7. Search for a word 8. Exit.

C Program

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

void combineNames(const char first[], const char middle[],
                  const char last[], char full[]) {
    strcpy(full, first);
    strcat(full, " ");
    strcat(full, middle);
    strcat(full, " ");
    strcat(full, last);
}

void findLength(const char full[], size_t *length) {
    *length = strlen(full);
}

void copyName(const char full[], char copied[]) {
    strcpy(copied, full);
}

void compareNames(const char first[], const char last[],
                  int *result) {
    *result = strcmp(first, last);
}

void reverseNameOrder(const char first[], const char middle[],
                      const char last[], char reverse[]) {
    strcpy(reverse, last);
    strcat(reverse, " ");
    strcat(reverse, middle);
    strcat(reverse, " ");
    strcat(reverse, first);
}

void countCharacters(const char full[], int *vowels,
                     int *consonants, int *digits, int *spaces) {
    int i;
    *vowels = *consonants = *digits = *spaces = 0;
    for (i = 0; full[i] != '\0'; i++) {
        unsigned char ch = (unsigned char)full[i];
        char c = (char)tolower(ch);
        if (isalpha(ch)) {
            if (c == 'a' || c == 'e' || c == 'i' ||
                c == 'o' || c == 'u')
                (*vowels)++;
            else
                (*consonants)++;
        } else if (isdigit(ch)) {
            (*digits)++;
        } else if (isspace(ch)) {
            (*spaces)++;
        }
    }
}
```

```
void searchWord(const char full[], const char word[], int *found) {
    *found = (strstr(full, word) != NULL);
}

int main(void) {
    char first[50], middle[50], last[50];
    char full[160], copied[160], reverse[160], word[50];
    int choice, result, v, c, d, s;
    size_t length;

    printf("Enter first name: ");
    scanf("%49s", first);
    printf("Enter middle name: ");
    scanf("%49s", middle);
    printf("Enter last name: ");
    scanf("%49s", last);
    combineNames(first, middle, last, full);

    do {
        printf("\n1. Full name\n2. Length\n3. Copy name\n");
        printf("4. Compare names\n5. Reverse order\n");
        printf("6. Count characters\n7. Search word\n8. Exit\n");
        printf("Enter choice: ");
        if (scanf("%d", &choice) != 1) {
            printf("Invalid input.\n");
            break;
        }
        switch (choice) {
            case 1:
                printf("Full name: %s\n", full);
                break;
            case 2:
                findLength(full, &length);
                printf("Length: %zu\n", length);
                break;
            case 3:
                copyName(full, copied);
                printf("Copied name: %s\n", copied);
                break;
            case 4:
                compareNames(first, last, &result);
                if (result == 0)
                    printf("First and last names are equal.\n");
                else
                    printf("First and last names are different.\n");
                break;
            case 5:
                reverseNameOrder(first, middle, last, reverse);
                printf("Reverse order: %s\n", reverse);
                break;
            case 6:
                countCharacters(full, &v, &c, &d, &s);
                printf("Vowels: %d\nConsonants: %d\n", v, c);
                printf("Digits: %d\nSpaces: %d\n", d, s);
                break;
            case 7:
                printf("Enter word to search: ");
                scanf("%49s", word);
                searchWord(full, word, &result);
                printf(result ? "Word found.\n" : "Word not found.\n");
                break;
            case 8:
                printf("Exiting program.\n");
                break;
            default:
                printf("Invalid choice.\n");
        }
    } while (choice != 8);
```

```
    return 0;
}
```

Sample Output (selected menu choices)

```
Enter first name: Tiksha
Enter middle name: Kumari
Enter last name: Sharma
Enter choice: 1
Full name: Tiksha Kumari Sharma
Enter choice: 2
Length: 20
Enter choice: 5
Reverse order: Sharma Kumari Tiksha
Enter choice: 7
Enter word to search: Kumari
Word found.
Enter choice: 8
Exiting program.
```

Result: The menu-driven program performs all seven string operations using separate user-defined functions.